This question paper contains 4+2 printed pages]

Roll No. | | | | | | | | | | | |

S. No. of Question Paper : **9438A**

Unique Paper Code : **32517919** **HC**

Name of the Paper : **Transmission Lines, Antenna and Wave Propagation**

Name of the Course : **B.Sc. (Hons.) Electronics : DSE-4**

Semester : **VI**

Duration : **3 Hours** Maximum Marks : **75**

(Write your Roll No. on the top immediately on receipt of this question paper.)

Attempt *five* questions in all, including

Question No. **1** which is compulsory.

All questions carry equal marks.

Use of Scientific Calculator is allowed.

1. Attempt any *five* :

(*a*) For silver ($\sigma = 6.1 \times 10^7$ mhos/m) and $\mu_r = 1$. Find the frequency at which the skin depth is 2 mm.

(*b*) A two-conductor transmission line supports TEM mode. Explain. 3

(*c*) What is a dominant TE mode in a rectangular wave guide and draw its field configuration. 3

(*d*) Define radiation pattern, directive gain and power gain of an antenna. 3

(*e*) A uniform plane wave propagating in a medium has :

$$E = 2e^{-\alpha z} \sin(10^8 t + \beta z) a_y \text{ V/m}$$

with $\varepsilon_r = 1$, $\mu_r = 20$ and $\sigma = 3$ mho/m. Find α and β. 3

(*f*) Explain the difference between the Broadside and End-fire array. 3

(*a*) For an Electromagnetic wave propagating in conducting medium, find the expressions for propagation constant, wave velocity and intrinsic impedance. 7

(*b*) Prove that attenuation and phase constants have equal magnitude for good conductors. 3

(*c*) A uniform plane wave in air with $E = 8 \cos(\omega t - 4x - 3z) a_y$ V/m is incident on a dielectric slab ($z \geq 0$) with $\varepsilon_r = 2.5$, $\mu_r = 1$. Find the polarization of wave and angle of incidence. 5

3. (*a*) Derive the transmission line equations. Also find the expressions for characteristic impedance of the line. 6

(*b*) Discuss open circuited, short circuited and matched transmission lines. 6

(*c*) An air-line has a characteristic impedance of 70 Ω and a phase constant of 3 rad/m at 100 MHz. Calculate the inductance per meter and the capacitance per meter of the line. 3

4. (*a*) What is a rectangular wave guide ? Derive expressions for Electric field, Magnetic field components and cut-off frequency in a rectangular wave guide for TM mode. 7

(*b*) Explain why a rectangular wave guide can't support TEM mode. 3

(*c*) A rectangular wave guide of dimensions 3 cm × 2 cm operates at 10 GHz. Find cut-off frequency, cut-off wavelength, free space wavelength, guided wavelength, guided phase constant and phase velocity of TE_{10}. 5

5. (*a*) What is a Hertzian dipole ? Derive an expression of radiation resistance for Hertzian dipole. 7

(*b*) What is a Yagi-Uda antenna ? Explain its working with the help of diagram. Write its one application. 4

(*c*) Find the maximum effective area of a half-wave dipole antenna operating at 30 MHz. How much power is received with an incident plane wave of strength 2 mV/m ? Given that maximum directive gain is 1.64. 4

6. (*a*) In normal incidence at the interface of a perfect dielectric to perfect conductor case, prove that total reflected wave combines with the incident wave to form a standing wave with electric and magnetic field components as : 7

$$E_1 = 2E_{io} \sin(\beta_1 z) \sin(\omega t) a_x \text{ V/m}$$

$$H_1 = \frac{2E_{io}}{\eta_1} \cos(\beta_1 z) \cos(\omega t) a_y \text{ V/m.}$$

(*b*) Discuss how a $\lambda/4$ line acts as an impedance transformer. What are its main disadvantages ? 4

(*c*) A lossless line having $Z_o = 50\ \Omega$ is 30 m long and operates at 2 MHz. The line is terminated with a load $Z_L = 60 + j40\ \Omega$. If $u = 0.6c$ (where c is the speed of light in a vacuum) on the line, find :

(*i*) The reflection coefficient Γ

(*ii*) The input impedance. 4

7. (*a*) What is a wave guide resonator and how the wave guide resonator is different from a conventional LC circuit ? 6

(*b*) What is a loop antenna ? Mention its *two* applications. 4

(*c*) Evaluate the directivity of an antenna with normalized radiation intensity : 5

$$U(\theta, \phi) = \begin{cases} \sin(\theta) & 0 \le \theta \le \pi/2,\ 0 \le \phi \le 2\pi \\ 0 & \text{otherwise} \end{cases}$$

Physical Constants :

$$\varepsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$$

$$c = 3 \times 10^{8} \text{ m/s.}$$